textbooks on the subject had been written before, but von Neumann breezily dismissed the best known and most technically sophisticated of them in the introduction to his own, claiming (accurately) that it "in no way satisfies the requirements of mathematical rigor." The book contained von Neumann's subtly flawed "impossibility proof," but this errant result was a (nearly invisible) blemish on what was otherwise an impressive technical achievement. Von Neumann articulated quantum physics in a mathematics as formal as his clothes, deriving well-known results from a handful of fundamental postulates. Among those postulates was one that von Neumann knew to be essential to the theory as it was understood at the time: he stated that wave functions normally obey the Schrödinger equation but collapse on measurement. "We therefore have two fundamentally different types of interventions which can occur in a system," von Neumann wrote. When an object remains undisturbed, the Schrödinger equation "describes how the system changes continuously and causally in the course of time." But, once a measurement is made, the smooth regularity of the Schrödinger equation goes out the window. "The arbitrary changes by measurements," said von Neumann, are "discontinuous, non-causal, and instantaneously acting."

Here, von Neumann dissented from Bohr's view. Bohr held that measurement devices and other large objects must be described in the language of classical physics, and that this somehow accounted for the results of quantum experiments without invoking any kind of wave function collapse. Exactly how this worked was something Bohr and his followers were all unclear about—and this lack of clarity was unacceptable to von Neumann in his quest to make quantum physics more mathematically rigorous. Instead, he held that quantum physics applied to large objects as well as small ones. Quantum physics, in von Neumann's view, was a theory of the whole world. But this made the measurement problem far more stark. If normal objects are subject to the laws of quantum physics in the same way that atoms are, then normal objects can't collapse wave functions, since wave function collapse violates the Schrödinger equation. And if normal objects don't collapse wave functions, that would lead straight to the paradox of Schrödinger's cat. The punk-rock particles from the Introduction were in two seemingly contradictory states—a bizarre situation known as a *superposition*—and because their wave function never collapsed, they ultimately forced Schrödinger's cat to be in a superposition as well,